Cisco CCNA 200-301 - The Complete Guide To Getting Certified

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Chapter 1

Host to host communications

1.0.1 A very basic introduction to networking

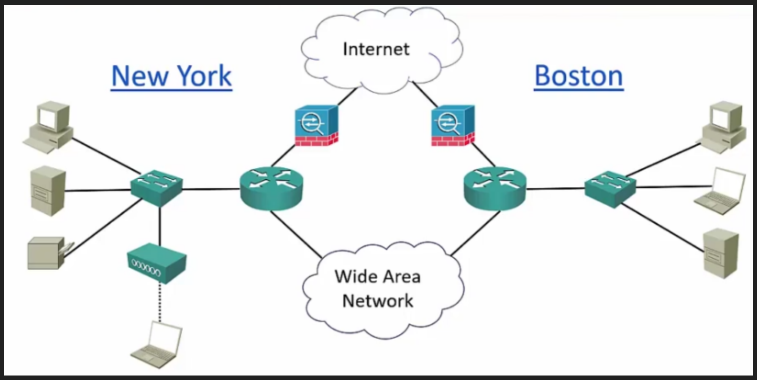
Let's say we have an office in NY. In that office there are end-hosts, such as PCs, server, printers, etc. All of the end hosts need to communicate with each other, for this we introduce a switch, this switch is going to be connected to each of the end-hosts using ethernet cables. A switch is a device with a lot of ports - which are basically slots to connect ethernet cables to - this allows the device to be connected to each other through the switch. The switch is what allows connectivity on my *local area network.* Let's say we add another end-host, a laptop, which is connected wirelessly to the same switch, the laptop will be connected via WIFI to a *wireless access point* and the wireless access point is connected to the switch via an ethernet cable. Now that all the end-hosts are able to communicate, we have built a *local area network*, which means a network that connects devices within the same local area, such as an office or a university campus.

Now that we have built a local area network, we also want these end-hosts to be able to communicate with other devices out on the internet, not just in my local area network. For this we introduce another component, the *router*, which is a device capable of making advanced routing decisions to route traffic between different areas of your network.

Next, now that we have connected the end-hosts to the local are network, and the local area network to the internet via a router. The internet is a very dangerous place, hackers can attack us and other things that we don't want can happen. For security reasons we install a *firewall*, which secures the different parts of our network from each other.

In a larger company we would not just have end-hosts connected to switches connected to routers connected to firewalls connected to the internet. We would have similar local area networks in many other places, for example, we can say that we have a very similar local area network in Boston. The same devices and the same connections. Lets say we want the NY and Boston office to communicate directly without using the internet. We can use a *dedicated connection* between my two routers on both sites for communications to be direct from the NY router to the Boston router, this is called a *wide area network.*

All this is expressed in the next, *network topology diagram.*



Some other caracteristics of networks are:

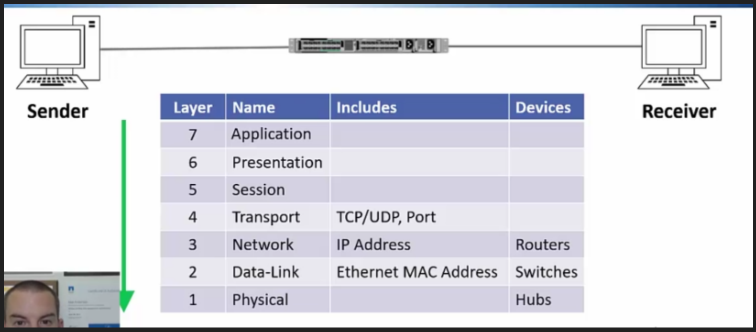
- Topology: how the devices are connected to the each other.
- Speed: the faster, the more cost.
- Cost: the monetary cost, effected by the type of devices and technologies used.
- Security: all the security features inside and outside the devices.
- Availability: making sure that our network always stays up. Mission critical components are typically duplicated, that way we can have an alternative if one were to fail.
- Scalability: the topology design lends itself to grow as needed.
- Reliability: similar to availability, making sure the network is going to continue to work.

1.1 The OSI Reference model

- The OSI model stands for Open Systems Interconnect model.
- It is standardized by the Organization for Standardization (ISO).
- It is a general purpose network that standardizes how computers communicate with one another over a network.
- The OSI model is conceptual it is not a physical thing, device, or protocol. It is a concept for humans only, to be able to understand.
- The 7 layer approach to data transmission divides the operations into specific related groups of actions at each layer. A layer serves the layer above it and is served by the layer below it.

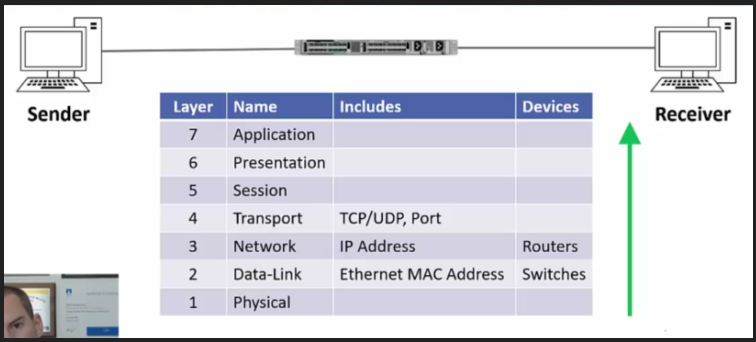
1.1.1 Example

An example is how you send an email. The sender (the person sending the email) and the receiver (the person receiving the email). In the layer 7 - the application layer - which is the layer containing the "from:" or the "to:" field, this is collected at an application layer. Next the data will be encapsulated by the layer 6 - presentation layer - which deals with the format the data is being handled with. Then the layer 6 information is encapsulated into the layer 5 - session layer - information. That then gets encapsulated to the 4th layer - the transport layer - which is the layer that specifies what communcation will be used - TCP, UDP, Port number - to prepare it for the next layer. That gets encapsulated into layer 3 - the network layer - which includes the source and destination IP address, a device that operates at layer 3 is the routers. The next the packet is almost being finished, and passed on to the layer 2 - the data link layer - which specifies the mac address - in case we are sending this over an ethernet network -, devices that operate at this layer are switches. Finally the encapsulated and consolidated packet is ready to be sent, this is done in layer 1 - which is the Physical layer - which literally means sending the packet through the wire, a network device that used to be used in this layer are hubs - which are not used anymore -. The email has been sent. The packets travel through the wires and devices to the receiver and the receiver receives packets in the form of electrical signals.



Layer	Name	Includes	Devices
7	Application		
6	Presentation		
5	Session		
4	Transport	TCP/UDP, Port	
3	Network	IP Address	Routers
2	Data-Link	Ethernet MAC Address	Switches
1	Physical		Hubs

The receiver will process the information being received from layer 1 to layer 7, which means that the receiver will grab the data from the first layer and decapsulate it until it has layer 7 data. It comes in at the physical layer, then it it will pass to the data link layer which is 2nd layer, it will perform checks to ensure that the the destination mac address is really for the receiver, if it is not, the receiver will just discard the packet. Next it will look at the layer 3 - which is the network layer - it will check the destination ip address and check if the packet is for it, otherwise it will discard the packet. Then we carry on with the next layer, layer 4 - the transport layer - it will check the port number in which it was received and some other validations. It will continue through the session layer - layer 5 -, the presentation layer - layer 6 -, and the application layer - layer 7 -.



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Layer 6 and 7 - application and presentation - are considered the upper level, typically used by developers not network engineers. Usually network engineers start to become interested from layer 4 downward.

1.1.2 Benefits of the OSI model

- Engineers don't need to design technologies to work end to end from top to bottom of the model. Instead they can just focus on their layer of expertise, and make sure they comply with the standards for the layers above and below.
- This leads to open standards and multi-vendor interoperability.
- For example: if you're an application developer, you can just focus on the top three layers, the lower layers are the domain of network engineers. You don't need to understand thzem, you just make sure you are complying with the standards.
- Troubleshooting is easier because you can analyse a problem in a logical fashion layer by layer.
- It is difficult to convey how important the OSI model is. The OSI model is used to describe problems, solutions, technologies, etcetera. Such as if the cable was disconnected you would say 'it was a layer 1 problem', if you find that the problem was that the user specified a wrong destination - you say - 'it was a layer 7 problem, etcetera.

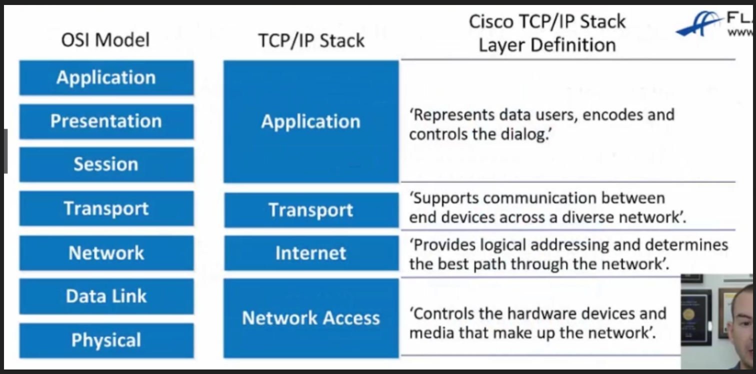
1.1.3 OSI Acronyms

- The classic: Please Do Not Throw Sausage Pizza Away.
- Network Relevant: Please Don't Need Those Stupid Packets Anyway.
- Us relevant: Please Do Not Teach Students Pointless Acronyms.
- Useful: Please Do Not Take Sales People's Advice.
- Favorite: Please Do Not Touch Superman's Private Area.

1.2 TCP/IP Stack

- TCP/IP was developed during the 1960s by the US department of defense's Advanced Research Projects Agency (ARPA).
- It is a protocol stack which consists of multiple protocols including TCP (Transmission Control Protocol) and IP (Internet Protocol). It is not a single protocol, it is a protocol stack.
- An analogy to understand what a protocol is is when diplomats meet each other, there is a certain way they are expected to behave and to communicate with each other, in computer networking protocols basically mean the same thing. You have two hosts and they are going to communicate with each other, there has to be a protocol that describes that communication is going to be and how is it going to behave.
- It is the main protocol stack used in computer operations today.
  - There used to be many other protocols such as Ipxspx and apple talk. But today TCP/IP is pretty much standard, other protocols are pretty much dead.
- Whereas the OSI reference model is conceptual, the TCP/IP stack is actually used to transfer data in production networks.
- TCP/IP is also layered but does not use all the OSI layers, though the layers are equivalent in operation and function.
  - It does actually use all of the layers, but in the documentations only 4 are layed out.

1.2.1 Comparing the OSI model with the TCP/IP



Notice that the TCP/IP does cover all the layers of the OSI model, however it sometimes uses the layers together. The application layer - in the TCP/IP - maps to the OSI model's layer 5,6, and 7 - session, presentation, and application -. Below that we have the transport layer - in the TCP/IP stack - which notice maps directly to the transport layer in the OSI model. The internet layer in the TCP/IP is equivalent to the network layer in the OSI model. Last we have the network access in the TCP/IP stack that maps to the OSI layer 1 and 2 - physical and data-link-.

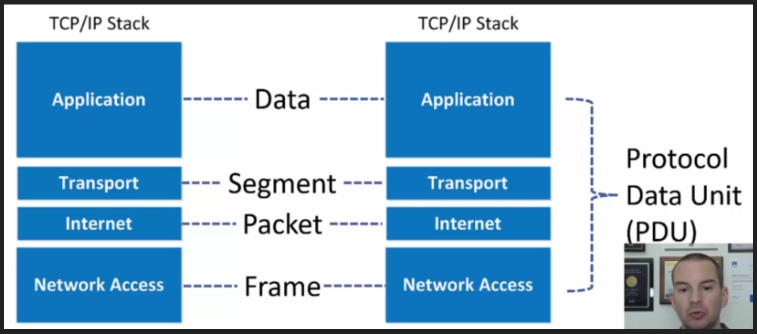
- While both the OSI model and the TCP/IP stack are very important in networking. We are going to typically be referencing more the OSI model than the TCP/IP stack.

1.2.2 Hosts communication terminology

When two hosts communicate with each other, they talk using something called protocol data unit (PDU), which is the entire communication that happens on the OSI model (sender from layer 7 to layer 1 and receiver from layer 1 back to layer 7).

The data exchanged varies in terminology according to which layer of the TCP/IP stack is being talked about.

- The application layer communicates using data to the application layer in the other host.
- The transport layer communicates using segments to the transport layer in the other host.
- The internet layer communicates using packets to the internet layer in the other host.
- The network access layer communicates using frames to the network access layer of the other host.



In the networking world the OSI model is more commonly referenced than the TCP/IP stack in terms of layers - which means that it is more common to refer to the layers of the OSI model than the layers of the TCP/IP -. However when we are talking about host communications terminology, it is more common to speak of the TCP/IP communications terminology - which means that it is more common to use 'data' if you are refering to the layer 5,6, and 7 of the OSI model, layer 3 call it a packet, layer 4 call it a segment-. Another very important thing is that networking people usually refer to the PDU as the name packet, the correct name is a PDU not a packet since the packet is used for layer 2 (internet layer) communication, but people typically use the term packet and PDU interchangeably, while technically wrong, but in day-to-day communcations when refering to host communications it is not uncommon to call it packet meaning the entire stack.

1.3 The upper OSI layers

- The networking engineers do not typically work directly with the upper 3 layers of the OSI model, but we still need to know what they do.
- They are more relevant to application developers.
- In this lecture I will primarily be giving you the Cisco definitions of the layers.
- Information included in the upper layers would include the message body and subject line in an email message for example.

1.3.1 Layer 7 - the application layer

- The application layer provides network services to the applications of the end user.
- If differs from the other layers in that it does not provide services to any other OSI layer.
- The application layer establishes the availability of intended communications partners.
  - Intended communcation partners are the hosts that the current hosts is trying to communicate with.
- It then synchronizes and establishes agreement on procedures for error recovery and control of data integrity.
  - Data integrity is verifying that data has not been corrupted or damaged in transit.

1.3.2 Layer 6 - the presentation layer

- The presentation layer ensures that the information that is sent at the application layer of one system is readable by the application layer of another system.
- The presentation layer can translate among multiple data formats using a common format (eg computers with different encoding schemes)

1.3.3 Layer 5 - session layer

- The session layer establishes, manages, and terminates sessions between two communicating hosts.
- THe session layer also synchronizes dialog between the presentation layers of the two hosts and manages their data exchange.
- For example, web servers have many users, so there are many communication processes open (session) at anygiven time to track. The session layer keeps track of all of the opened sessions.
- It also offers efficient data transfers, CoS (Class of Service), and exception reporting of upper layer problems.

1.4 The lower OSI layers

- Whereas network engineers are not particularly interested in the upper OSI layers, we are very concerned with the lower 4 layers of the OSI model. These are incredibly important and are almost like the daily routine to network engineer.
- Each of these layers have their own dedicated section later and you will learn much more detailed information about them throughout the course.

1.4.1 Layer 4 - transport layer

- The main characteristics of the transport layer are wether TCP or UDP transport is used, and the port number.
- We haven't really seen what is TCP and UDP, for now you can think of it as modes of transportation where each one has different advantages. The TCP is very reliable, therefore if priority is reliability we use TCP. If speed is more important than reliability - like for voice of video traffic - then we use UDP instead.
- We haven't seen what ports are, but for now you just need to know that different processes occur in these ports, for example the 80th port is used for http web traffic and port 25 is used for email.

- Definition:
  - The transport layer defines services to segment, transfer, and reassemble the data for individual communications between the end devices.
  - It breaks down large files into smaller segments that are less likely to incur transmission problems.

1.4.2 Layer 3 - network layer

- The most important information at the network layer is the source and destination IP address. Other information is also carried, but the main is the source and destination of IPs.
- Routers operate at layer 3.
- Definition:
  - The network layer provides connectivity and path selection between two host systems that may be located on geographically separated networks.
  - The network layer is the layer that manages the connectivity of hosts by providing logical addressing (IP addressing is our logical addressing).

1.4.3 Layer 3 - the data-link layer

- The most important information at the data-link layer is the source and destination layer 2 address. Again - just like in layer 3 and 4 - there is also other information but the main one is the source and destination layer address.
- For example the source and destination MAC address if ethernet is the layer 2 technology. Different layer 2 technologies use different formats for their addressing, for example, old legacy frame relay uses DLCI or DCE numbers for the addressing. Today the typical address used at this layer is the MAC address.
- Switches operate at layer 2.
- Definition:
  - The data link layer defines how data is formated for transmission and how access to physical media is controlled.
  - It also typically includes error detection and correction to ensure a reliable delivery of data.

1.4.4 Layer 1 - the physical layer

- The physical layer concerns literally the physical components of the network, for example the cables being used.
- Definition:
  - The physical link enables bit transmission between end devices.
  - It defines specifications needed for activating, maintaining, and deactivating the physical link between end devices.
  - For example, voltage levels, physical data rates, maximum transmission distances, physical connectors, etcetera.

Chapter 2

The cisco IOS Operating System

2.1 Introduction

By now we all want to configure the cisco devices such as routers and switches. Cisco devices do include GUIs - which means graphical user interface - but in the real world, everybody just uses the command line interface (CLI) because it is much quicker than using the GUIs. The OS used in enterprise grid cisco devices is IOS, in this section we'll learn how to connect to the devices and the use of the command line. A disclaimer is that the IOS operating system is not the only one, there are others.

2.2 Cisco operating systems

Cisco history is not going to be examined on the CCNA exam. Neil includes this just because of context. Cisco IOS and the other OS's work basically the same way in terms of management and administrative point of view, the operating systems don't differ that much in terms of how you use it, maybe on how they are actually implemented but not really on how you use it. By learning cisco IOS you are really learning all of the others at once.

2.2.1 A short history of cisco operating systems

- Most people think of cisco as primarily a routing and switching company, but they actually started out with just routers in 1984.
- IOS was the original operating system that they used on routers - the same OS they use today (with some optimizations).
- IOS is the operating system that has been used on cisco routers since their inception.
- Cisco Catalyst switches evolved from the acquisition of Crescendo company in 1993 - cisco bought the company Crescendo- and started manufacturing switches of which the first one was Catalyst.
- The original cisco switch operating system was CatOS which has not been deprecated.
- Cisco firewalls evolved from the acquisition of the company Network Translation's PIX firewall that uses the Finesse operating system in 1995.
- Cisco switches and firewalls (now the ASA firewall) were ported over the IOS operating system over the following years.
- Cisco standardized on IOS for all the network infrastructure devices.

There are some other operating systems in some newer router and switch platform. IOS runs on the majority of routers and switches. Some of these other operating systems are:

- The Cisco Nexus and MDS data center switch product lines run on NX-OS.
- The IOS-XR operating system runs on the service provider NCS, CRS, ASR9000, and XR12000 series routers (these are the higher end routers).
- IOS-XE runs on the ASR1000 series service provider routers (1K service providers).
- The command line interfaces for teh other operating systems are nearly identical to IOS.

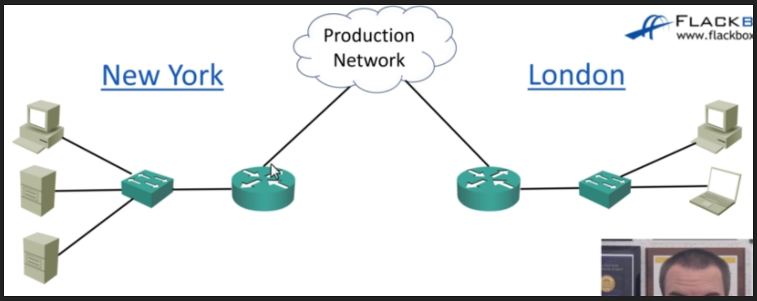
At first all of these names might seem intimidating but fear not, all of the OS's run nearly identical commands, if you can use one, you can practically use them all. There are different OS's because they are implemented with different functionality, for example, IOS has a monolithic kernel - which means that if one process crashes it crashes the entire router - they are all different under the hood. Other OS's have different kernels to permit that if one crashes the other processes in the router are not affected, only that process is crashed. This is does not mean that IOS is bad or unreliable, it simply means that it works differently given different situations.

2.3 Connecting to a cisco device over the network

- The lab exercises in this course use Cisco Packet Tracer simulation software on your PC.
- This lecture shows how to connect to a real router or switch over the network with Putty - which a software to connect to routers - you are not going to use Putty in the labs because everything is done in the Packet Tracer simulation software.
- You do not need to install or use Putty to do the course lab exercises.

2.3.1 Connecting over the network

In you day-to-day as a network engineer you are typically not going to be connecting stuff, instead you are going to be in your desk 'working remotely', you are going to be connecting to that device over the network. So the following example: Let's say that you are in a London office and want to establish a connection in the NY office. What you are going to do is use software on your PC, connect to the router over the network in NY, get to the command line of device you want to work on, and get on with your job.

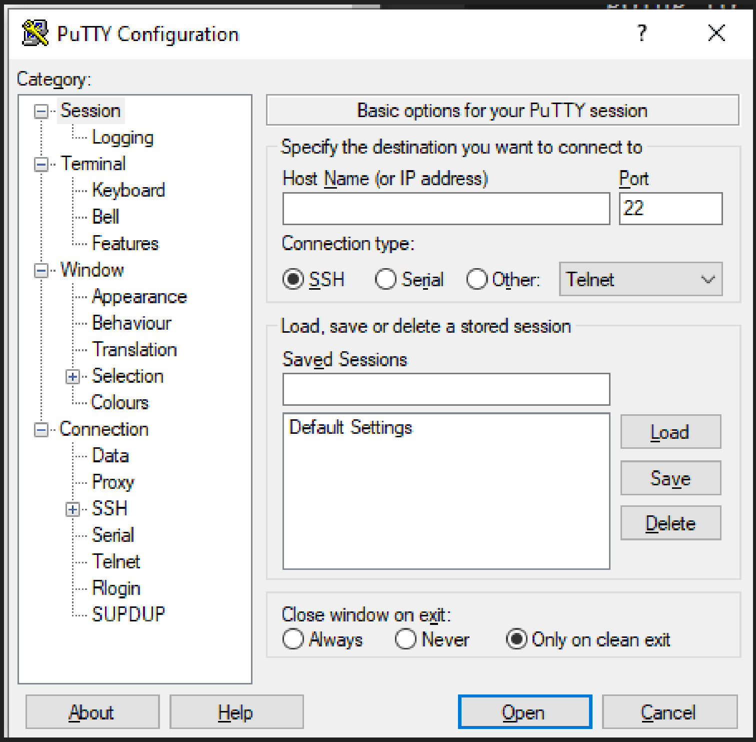


- To get to the CLI for day-to-day management of a Cisco device you will use Secure Shell (SSH) to connect to it's management IP addresses over the network.
- There is also another protocol called Telnet which is also supported but not recommended because it is insecure. Commands in Telnet are un-encrypted, so anybody listening in to your traffic can get ahold of data, including username and password. Telnet is never used in a real setting for the potential vulnerabilities. In reality SSH is always used, Telnet is almost always used only on a lab environment to learn.
- In enterprise networks, secure login will typically be enforced through intergration with a centralized AAA (Authentication, Authorization, and Accounting) server.

– Basically the AAA server is a solution to many people loggin in. In a network, you can probably have many devices with username and passwords that must be verified to be granted access to network actions. We don't want each device to have the username and passwords of all the others so we centralize that into one server that stores all the user name and passwords that has access to the central database of usernames and passwords.

2.3.2 Installing Putty

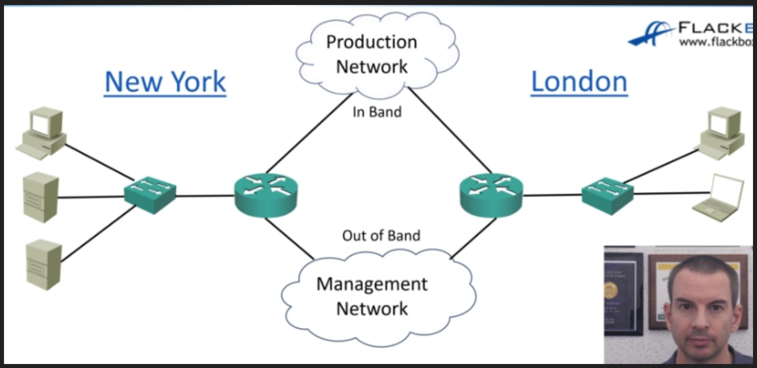
Installing Putty is basically just searching on Google and accepting all the defaults once the executable has been downloaded. When you launch the window you will be faced with a window like this:



The host name refers to the host that I am intending to connect to, it takes as an input an IP address. Then you log in through the command line, and type your password. This is how you connect to a cisco device, very simple using Putty.

2.3.3 Out of band management

To continue the example in which we were connecting in London to the NY office, we are essentially using the production network. But most large companies use a separate management network, this is done as a backup path in case the production network is down, it is also more secure to have a separate dedicated path. The production network is the in band network, which means you are using the same cables that your normal users (your staff) are using to connect to other devices. If you connect to the NY office through the management network you are using and out of band network, which means you are not using the same network as the normal staff. Trivial difference but important distinction.


Production
Network
New York
London
In Band
Out of Band
Management
Network